# IGCSE 0607 Practice Questions — E4.6 — Combined Questions & Markscheme

## Perpendicular lines

**Total Questions:** 6 — **Total Marks:** 36 Marks

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- **E4.6** Perpendicular lines
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

**1** Find the equation of the line that is perpendicular to $y = 3x - 5$ and passes through the point $(6, 2)$. Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**2** Line $L_1$ has the equation $y = 5 - 2x$.
Line $L_2$ is perpendicular to $L_1$ and passes through the point $(4, 1)$.

**(a)** Find the gradient of line $L_1$.

..... **[1]**

**(b)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

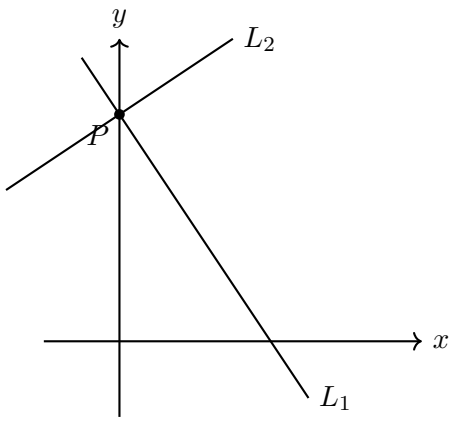
$y =$ ..... **[3]**

**3** $A$ is the point $(-4, 7)$ and $B$ is the point $(2, -1)$.

Find the equation of the perpendicular bisector of the line segment $AB$.
Give your answer in the form $ax + by = c$, where $a$, $b$ and $c$ are integers.

..... **[5]**

**4** Line $L_1$ has the equation $3x + 2y = 12$.
Line $L_2$ is perpendicular to $L_1$ and intersects $L_1$ at a point $P$ on the $y$-axis.




*Not to scale*

**(a)** Find the coordinates of point $P$.

( ..... , ..... ) **[1]**

**(b)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(c)** Point $Q$ lies on $L_2$ and has an $x$-coordinate of 9.
Point $R$ lies on $L_1$ and has an $x$-coordinate of 4.
Work out the area of triangle $PQR$.

..... **[5]**

**5** The points $A$ and $B$ have coordinates $(-2, 4)$ and $(6, 0)$ respectively.

**(a)** Find the coordinates of the midpoint of $AB$.

( ..... , ..... ) **[2]**

**(b)** Find the equation of the perpendicular bisector of the line segment $AB$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[4]**

**6** The line $L_1$ has equation $2y + 3x = 12$.
The line $L_2$ is perpendicular to $L_1$ and intersects $L_1$ at the point $P$ on the $y$-axis.

**(a)** Find the coordinates of $P$.

( ..... , ..... ) **[2]**

**(b)** Find the equation of line $L_2$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[3]**

**(c)** Line $L_2$ crosses the $x$-axis at the point $Q$.
Calculate the exact length of $PQ$.

..... **[4]**

**1** Identifies the gradient of the given line as 3 **[B1]**

Uses the perpendicular gradient $-\frac{1}{3}$ and substitutes $(6, 2)$ into $y = mx + c$ or $y - y_1 = m(x - x_1)$ **[M1]**

$-\frac{1}{3}x + 4$ **[A1]**

[Back to Question] *[Total Marks: 3]*

**2 (a)** $-2$ **[B1]**

**(b)** Gradient of $L_2 = \frac{1}{2}$ **[M1]**

$1 = \frac{1}{2}(4) + c$ or $y - 1 = \frac{1}{2}(x - 4)$ **[M1]**

$y = \frac{1}{2}x - 1$ **[A1]**

[Back to Question] *[Total Marks: 4]*

**3** Calculates the midpoint of $AB = \left(\frac{-4+2}{2}, \frac{7+(-1)}{2}\right) = (-1, 3)$ **[B1]**

Calculates the gradient of $AB = \frac{-1-7}{2-(-4)} = -\frac{8}{6} = -\frac{4}{3}$ **[B1]**

Finds the perpendicular gradient $= \frac{3}{4}$ **[M1]**

Substitutes the midpoint $(-1, 3)$ and gradient $\frac{3}{4}$ to find the equation, e.g., $y - 3 = \frac{3}{4}(x - (-1))$ **[M1]**

$3x - 4y = -15$ or $4y - 3x = 15$ or equivalent integer form **[A1]**

[Back to Question] *[Total Marks: 5]*

**4 (a)** $(0, 6)$ **[B1]**

**(b)** Gradient of $L_1 = -\frac{3}{2}$ **[M1]**

Gradient of $L_2 = \frac{2}{3}$ **[M1]**

$y = \frac{2}{3}x + 6$ **[A1]**

**(c)** Coordinates of $Q$: $y = \frac{2}{3}(9) + 6 = 12 \implies Q(9, 12)$ **[B1]**

Coordinates of $R$: $3(4) + 2y = 12 \implies y = 0 \implies R(4, 0)$ **[B1]**

Identify triangle $PQR$ is right-angled at $P$ OR setup valid box/shoelace method **[M1]**

Length $PQ = \sqrt{9^2 + 6^2} = \sqrt{117}$ and Length $PR = \sqrt{4^2 + (-6)^2} = \sqrt{52}$ **[M1]**

Area $= \frac{1}{2} \times \sqrt{117} \times \sqrt{52} = 39$ **[A1]**

[Back to Question] *[Total Marks: 9]*

**5** **(a)** $\left(\frac{-2+6}{2}, \frac{4+0}{2}\right)$ **[M1]**

$(2, 2)$ **[A1]**

**(b)** Gradient of $AB = \frac{0-4}{6-(-2)} = -\frac{1}{2}$ **[M1]**

Gradient of perpendicular bisector $= 2$ **[M1]**

Substitute midpoint $(2, 2)$ into $y - y_1 = m(x - x_1)$: $y - 2 = 2(x - 2)$ **[M1]**

$y = 2x - 2$ **[A1]**

[Back to Question] *[Total Marks: 6]*

**6** **(a)** Substitutes $x = 0$ into $2y + 3x = 12$ because $P$ is on the $y$-axis **[M1]**

$(0, 6)$ **[A1]**

**(b)** Rearranges $L_1$ equation to find gradient: $2y = -3x + 12 \implies y = -\frac{3}{2}x + 6$, so $m = -\frac{3}{2}$ **[B1]**

States perpendicular gradient for $L_2$ is $\frac{2}{3}$ **[M1]**

$\frac{2}{3}x + 6$ **[A1]**

**(c)** Substitutes $y = 0$ into the equation of $L_2$ to find the coordinates of $Q$ **[M1]**

$0 = \frac{2}{3}x + 6 \implies x = -9$ **[A1]**

Uses the distance formula for $P(0, 6)$ and $Q(-9, 0)$: $\sqrt{(-9-0)^2 + (0-6)^2}$ **[M1]**

$\sqrt{117}$ or $3\sqrt{13}$ **[A1]**

[Back to Question] *[Total Marks: 9]*